

CTMT Covariance Geometry: Resolution-Relative Covariance, Null Support, Coupling Invariants, Similarity, Falsification, and Non-Claims

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Abstract

This paper develops the covariance-geometry layer of Chronotopic Metric Theory (CTMT). It complements the local resolution-geometry formulation in which an operational chart determines a resolved sector, an unresolved sector, and an observational covariance structure

$$\mathcal{G} = (O; R, N, \Sigma).$$

The purpose of the present paper is not to introduce a new universal covariance estimator, nor to claim that the Jacobian alone determines the complete covariance matrix. Rather, the purpose is to identify what CTMT contributes: a resolution-relative geometry of covariance.

Given a finite-dimensional observation space O , an observable-response Jacobian $J : T_\theta\Theta \rightarrow O$, and an observational covariance Σ , the Jacobian determines the resolved and unresolved sectors

$$R = \text{Im } J, \quad N = \ker J^\top,$$

subject to the declared observation pairing. Relative to the decomposition $O = R \oplus N$, every covariance admits a block representation

$$\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^\top & \Sigma_N \end{pmatrix}.$$

We call the tuple

$$\text{CovGeom}(J, \Sigma) := (R, N, \Sigma_R, \Sigma_N, C_{RN})$$

the CTMT covariance geometry of the chart. The components have distinct operational meanings: Σ_R is resolved covariance, Σ_N is null covariance, and C_{RN} is resolved–null covariance coupling. The cross block is not a nuisance term. It changes the stabilizer group of the chart and therefore changes the admissible local morphism class.

The main theorem states that strict admissible chart isomorphisms preserve this covariance geometry, while conformal chart isomorphisms preserve it up to one common positive scale when and only when the protocol quotients out absolute covariance scale. Fisher geometry is recovered as a derived diagnostic,

$$G_F = J^\top \Sigma^{-1} J,$$

but is not sufficient to determine covariance geometry, since it may miss null-sector and resolved–null coupling defects. The paper defines resolution-relative similarity diagnostics, admissibility gates, covariance falsifiers, and conservative estimation protocol choices. The final claim is limited: CTMT does not replace established statistical covariance estimation; it supplies the geometric structure in which a declared covariance estimate must be decomposed, transported, compared, and falsified.

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1 Purpose and claim stratification

The local CTMT morphism problem is most conservatively formulated by the resolution geometry

$$\mathcal{G} = (O; R, N, \Sigma),$$

where O is an observation space, R is the resolved observable-response sector, N is the unresolved or null sector, and Σ is the covariance structure induced or declared by the measurement protocol.

The present paper addresses a different but adjacent question:

What exactly is covariance as a CTMT geometric object once the resolved/null decomposition has been fixed?

The answer proposed here is not that CTMT has a privileged universal estimator for Σ . Standard statistics already supplies many covariance estimators, including sample covariance, regularized covariance, robust covariance, state-space covariance, Bayesian covariance, and domain-specific instrumental covariance models. CTMT enters after the covariance is declared or estimated: it decomposes covariance relative to the resolution structure and tests whether the resulting geometry is admissible under transport.

The paper separates four layers.

Statistical layer. A covariance estimate or covariance model Σ is produced by a declared measurement protocol.

Resolution layer. A local observable-response Jacobian J determines

$$R = \text{Im } J, \quad N = \ker J^\top.$$

Covariance-geometry layer. Relative to $O = R \oplus N$, the covariance decomposes into

$$\Sigma_R, \quad \Sigma_N, \quad C_{RN}.$$

Morphism layer. Admissible local CTMT morphisms are precisely those maps that preserve the declared resolution geometry, strictly or conformally depending on the measurement protocol.

Remark 1 (Role of this paper). This paper is not a replacement for the automorphism classification of resolution geometries. It is the missing intermediate layer explaining how the covariance component of $\mathcal{G} = (O; R, N, \Sigma)$ should be decomposed, interpreted, compared, and falsified.

2 Operational observation chart

Let O be a finite-dimensional real observation vector space. A local measurement campaign supplies observations $y_k \in O$ and a declared map from admissible perturbation variables into observables.

Definition 1 (Operational observation chart). An operational observation chart is a quadruple

$$\mathcal{C} = (O, \Theta, F, \Sigma),$$

where:

- (a) O is the finite-dimensional observation space;
- (b) Θ is the finite-dimensional space of admissible local perturbation variables;
- (c) $F : \Theta \rightarrow O$ is the observable-response map;
- (d) Σ is the covariance structure supplied by the measurement protocol, estimated from data, or specified by an instrumental model.

Definition 2 (Observable-response Jacobian). At a base point $\theta_0 \in \Theta$, the observable-response Jacobian is

$$J = D_{\theta}F(\theta_0) : T_{\theta_0}\Theta \rightarrow O.$$

In applications, J may be computed by finite differences, automatic differentiation, local regression, inverse reconstruction, or another declared estimator. The computation of J is not merely numerical bookkeeping: rank stability and conditioning are admissibility data.

3 Resolved and unresolved sectors

Definition 3 (Resolved sector). The resolved sector is the first-order observable-response image

$$R := \text{Im } J \subset O.$$

Definition 4 (Null sector). Given the declared observation pairing used to represent sensor coordinates, the unresolved or null sector is

$$N := \ker J^{\top} \subset O.$$

When the pairing is Euclidean, $N = R^{\perp}$ and one has an orthogonal direct sum

$$O = R \oplus N.$$

More generally, the same statements hold after replacing orthogonality by the declared observation pairing.

Remark 2 (Null does not mean physically absent). The null sector is not a claim that the corresponding observable directions do not exist. It means that they are not activated by the declared admissible perturbations at first order. Null-sector covariance may therefore be operationally meaningful even when the Fisher pullback does not see it.

Definition 5 (Resolution projectors). Let Π_R and Π_N denote the projectors onto R and N associated with the declared splitting $O = R \oplus N$. In the Euclidean representation with Moore–Penrose pseudoinverse,

$$\Pi_R = JJ^+, \quad \Pi_N = I - JJ^+.$$

4 Covariance as a resolution-relative object

The covariance matrix Σ alone is not yet the CTMT covariance geometry. CTMT covariance geometry is the covariance seen through the resolved/null structure.

Definition 6 (Strict covariance structure). A strict covariance structure on O is a symmetric positive semidefinite bilinear form

$$\Sigma \in \text{PSD}(O)$$

whose absolute scale is part of the measurement protocol. When Σ is positive definite we write $\Sigma \in \text{SPD}(O)$.

Definition 7 (Conformal covariance structure). A conformal covariance structure is an equivalence class

$$[\Sigma] = \{\lambda\Sigma : \lambda > 0\},$$

used only when the observation protocol explicitly quotients out global covariance scale.

Definition 8 (Resolution-relative covariance blocks). Assume $O = R \oplus N$ and let $Q = [Q_R, Q_N]$ be an adapted basis. In this basis,

$$Q^\top \Sigma Q = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^\top & \Sigma_N \end{pmatrix},$$

where

$$\Sigma_R = Q_R^\top \Sigma Q_R, \quad \Sigma_N = Q_N^\top \Sigma Q_N, \quad C_{RN} = Q_R^\top \Sigma Q_N.$$

Definition 9 (CTMT covariance geometry). The CTMT covariance geometry of a chart is

$$\text{CovGeom}(J, \Sigma) := (R, N, \Sigma_R, \Sigma_N, C_{RN}).$$

In the conformal case it is

$$\text{CovGeom}_c(J, [\Sigma]) := (R, N, [\Sigma_R, \Sigma_N, C_{RN}]),$$

where the brackets denote one common positive scale on all covariance blocks.

The three covariance blocks have different meanings.

1. Σ_R is resolved covariance.
2. Σ_N is null covariance.
3. C_{RN} is resolved–null covariance coupling.

The cross block C_{RN} is especially important. If $C_{RN} = 0$, the resolved and null covariance geometries decouple. If $C_{RN} \neq 0$, admissible automorphisms must stabilize this coupling, and the morphism group generally shrinks.

5 Canonical block decomposition

Proposition 10 (Existence and uniqueness of block decomposition). Let $O = R \oplus N$ be a fixed decomposition and let Σ be a covariance form on O . Then Σ has a unique block decomposition

$$\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^\top & \Sigma_N \end{pmatrix}$$

relative to the chosen decomposition.

Proof 11. Every vector $x \in O$ decomposes uniquely as $x = x_R + x_N$ with $x_R \in R$ and $x_N \in N$. The bilinear form Σ is therefore uniquely determined by its restrictions to $R \times R$, $N \times N$, and $R \times N$. These restrictions are exactly Σ_R , Σ_N , and C_{RN} .

Corollary 12 (Covariance geometry is not Fisher geometry). The Fisher pullback

$$G_F = J^\top \Sigma^{-1} J$$

does not determine the full covariance geometry

$$(R, N, \Sigma_R, \Sigma_N, C_{RN}).$$

Proof 13. The Fisher pullback sees covariance only through the resolved response directions $Jv \in R$. Covariance changes supported entirely in N , or changes in resolved-null coupling that do not alter the pullback, may leave G_F unchanged while changing $\text{CovGeom}(J, \Sigma)$.

6 Null-supported covariance and seepage

The previous section treats the full observational covariance. A special case arises when one studies unresolved or seepage covariance.

Axiom 14 (Null admissibility of unresolved transport). An unresolved perturbation $u \in O$ is null-admissible if

$$J^\top u = 0.$$

Equivalently, $u \in N$.

Theorem 1 (Canonical support of null-admissible covariance). Assume unresolved perturbations satisfy $J^\top u = 0$ almost surely. Then their covariance $C_{\text{null}} = \text{Cov}(u)$ satisfies

$$C_{\text{null}} = \Pi_N C_{\text{null}} \Pi_N.$$

In particular,

$$\text{range}(C_{\text{null}}) \subseteq N.$$

Proof 15. If $u \in N$ almost surely, then $\Pi_N u = u$ almost surely. Hence

$$C_{\text{null}} = \mathbb{E}[(u - \mathbb{E}u)(u - \mathbb{E}u)^\top] = \Pi_N C_{\text{null}} \Pi_N.$$

Corollary 16 (Jacobian-induced null support). In the Euclidean representation,

$$\Pi_N = I - JJ^+$$

is the unique orthogonal projector onto the admissible null support.

Remark 3 (What the Jacobian determines). The Jacobian determines the support of null-admissible covariance. It does not determine the full covariance spectrum inside that support. The spectrum requires residual second moments, sensor-noise information, or another declared statistical model.

7 Transport of covariance geometry

Let two operational charts be given:

$$\mathcal{C} = (O, \Theta, F, \Sigma), \quad \mathcal{C}' = (O', \Theta', F', \Sigma').$$

Let

$$J = D_\Theta F(\theta_0), \quad J' = D_{\Theta'} F'(\theta'_0).$$

Definition 17 (Strict chart isomorphism). A strict local chart isomorphism consists of invertible maps

$$\phi : O \rightarrow O', \quad a : T_{\theta_0} \Theta \rightarrow T_{\theta'_0} \Theta',$$

such that

$$J'a = \phi J$$

and

$$\phi \Sigma \phi^\top = \Sigma'.$$

Definition 18 (Conformal chart isomorphism). A conformal local chart isomorphism satisfies the same intertwining relation

$$J'a = \phi J,$$

but covariance is preserved only up to one common positive scale:

$$\phi \Sigma \phi^\top = \lambda \Sigma', \quad \lambda > 0.$$

This is admissible only when the protocol explicitly discards global covariance scale.

Theorem 2 (Functoriality of covariance geometry). Let ϕ be a strict chart isomorphism. Then

$$\phi(R) = R', \quad \phi(N) = N',$$

and the covariance blocks transform as

$$\Sigma_R \mapsto \Sigma'_R, \quad \Sigma_N \mapsto \Sigma'_N, \quad C_{RN} \mapsto C'_{RN}.$$

Thus strict chart isomorphisms preserve CTMT covariance geometry.

Proof 19. The relation $J'a = \phi J$ with a invertible implies

$$\text{Im } J' = \phi(\text{Im } J),$$

so $\phi(R) = R'$. Preservation of the visible/invisible operational distinction gives $\phi(N) = N'$. Since $\phi \Sigma \phi^\top = \Sigma'$, restricting this identity to $R' \times R'$, $N' \times N'$, and $R' \times N'$ gives preservation of the three covariance blocks.

Corollary 20 (Conformal covariance geometry). Under a conformal chart isomorphism,

$$\Sigma_R, \quad \Sigma_N, \quad C_{RN}$$

are all multiplied by the same positive scale λ . Independent sector rescalings are not conformal CTMT covariance equivalences unless a separate sector-relative protocol is explicitly declared.

8 Relation to automorphism groups

For a single chart, let

$$\mathcal{G} = (O; R, N, \Sigma)$$

be its strict resolution geometry.

Definition 21 (Strict covariance-geometry automorphism). A strict covariance-geometry automorphism is an invertible map $\phi : O \rightarrow O$ such that

$$\phi(R) = R, \quad \phi(N) = N, \quad \phi \Sigma \phi^\top = \Sigma.$$

In an adapted decomposition $O = R \oplus N$, any such ϕ has block form

$$\phi = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}.$$

Substitution gives

$$A \Sigma_R A^\top = \Sigma_R,$$

$$B \Sigma_N B^\top = \Sigma_N,$$

and

$$A C_{RN} B^\top = C_{RN}.$$

Theorem 3 (Automorphisms are stabilizers of covariance geometry). The strict automorphism group of the local CTMT resolution geometry is

$$\text{Aut}(\mathcal{G}) = \left\{ \text{diag}(A, B) : A \Sigma_R A^\top = \Sigma_R, B \Sigma_N B^\top = \Sigma_N, A C_{RN} B^\top = C_{RN} \right\}.$$

Thus the automorphism group depends exactly on the CTMT covariance geometry.

Corollary 22 (Decoupled case). If $C_{RN} = 0$, then

$$\text{Aut}(\mathcal{G}) \cong O(\Sigma_R) \times O(\Sigma_N).$$

After whitening,

$$\text{Aut}(\mathcal{G}) \cong O(r) \times O(n - r),$$

where $r = \text{rank } J$ and $n = \dim O$.

Corollary 23 (Coupled case). If $C_{RN} \neq 0$, then $\text{Aut}(\mathcal{G})$ is the stabilizer of C_{RN} inside

$$O(\Sigma_R) \times O(\Sigma_N).$$

Therefore resolved-null covariance coupling is a morphism-class invariant.

Theorem 4 (Conformal automorphism group). For the conformal geometry $(O; R, N, [\Sigma])$,

$$\text{Aut}(\mathcal{G}_c) = \left\{ \text{diag}(A, B) : \exists \lambda > 0 \text{ such that } A \Sigma_R A^\top = \lambda \Sigma_R, B \Sigma_N B^\top = \lambda \Sigma_N, A C_{RN} B^\top = \lambda C_{RN} \right\}.$$

If $C_{RN} = 0$, then

$$\text{Aut}(\mathcal{G}_c) \cong \mathbb{R}_+ \times O(r) \times O(n - r).$$

9 Similarity of CTMT covariance geometries

The covariance-geometry viewpoint changes what it means for two systems to be similar. Similarity is not equality of Fisher metrics alone, nor equality of raw covariance matrices alone. It is similarity of covariance relative to the resolved/null geometry.

Let two charts produce

$$\text{CovGeom}_1 = (R_1, N_1, \Sigma_{R,1}, \Sigma_{N,1}, C_1),$$

$$\text{CovGeom}_2 = (R_2, N_2, \Sigma_{R,2}, \Sigma_{N,2}, C_2).$$

Definition 24 (Subspace distance). For subspaces of equal dimension, define

$$d_R = \|\Pi_{R_1} - \Pi_{R_2}\|_F, \quad d_N = \|\Pi_{N_1} - \Pi_{N_2}\|_F.$$

When covariance blocks are positive definite, one may use the affine-invariant SPD distance

$$d_{\text{SPD}}(A, B) = \left\| \log \left(A^{-1/2} B A^{-1/2} \right) \right\|_F.$$

For positive semidefinite or regularized blocks, one uses a declared regularized version.

Definition 25 (Resolution-relative covariance distance). A conservative CTMT covariance-geometry distance is

$$\begin{aligned} d_{\text{CTMT}} := & \alpha_R d_R + \alpha_N d_N \\ & + \beta_R d_{\text{SPD}}(\Sigma_{R,1}, \Sigma_{R,2}) + \beta_N d_{\text{SPD}}(\Sigma_{N,1}, \Sigma_{N,2}) \\ & + \gamma \frac{\|C_1 - C_2\|_F}{\max(\|\Sigma_1\|_F, \|\Sigma_2\|_F, \varepsilon)}. \end{aligned}$$

The weights must be declared by protocol and not tuned after seeing the desired conclusion.

Remark 4 (Similarity is protocol-relative). There is no universal scalar similarity for all CTMT systems. A coil test, a Navier window, a seismic chart, and a turbine telemetry chart may require different tolerances and normalizations. What is universal is the structure being compared:

$$R, \quad N, \quad \Sigma_R, \quad \Sigma_N, \quad C_{RN}.$$

10 Estimation protocol

The conservative CTMT covariance protocol is as follows.

Step 1. Declare observation space. Choose the observation vector $y \in O$ and preprocessing pipeline.

Step 2. Estimate or declare covariance. Use an appropriate established method to obtain $\hat{\Sigma}$. Examples include sample covariance, shrinkage covariance, robust covariance, instrumental noise covariance, or domain-specific state-space covariance.

Step 3. Compute the Jacobian. Estimate

$$J = D_\theta F(\theta_0).$$

Step 4. Report rank stability. Determine

$$r = \text{rank } J$$

under declared numerical tolerance. Report conditioning and sensitivity to the rank threshold.

Step 5. Compute resolved and null sectors. Compute bases for

$$R = \text{Im } J, \quad N = \ker J^\top.$$

Step 6. Block-decompose covariance. In an adapted basis, compute

$$\hat{\Sigma} = \begin{pmatrix} \hat{\Sigma}_R & \hat{C}_{RN} \\ \hat{C}_{RN}^\top & \hat{\Sigma}_N \end{pmatrix}.$$

Step 7. Report coupling strength. A useful normalized diagnostic is

$$\kappa_{RN} = \frac{\|\hat{C}_{RN}\|_F}{\|\hat{\Sigma}\|_F + \varepsilon}.$$

Step 8. Select strict or conformal interpretation. Use strict covariance geometry unless the protocol explicitly declares absolute covariance scale unobservable. In the conformal case, only one global scale is quotiented out.

Step 9. Test candidate morphisms. For a candidate ϕ , compute

$$d_R = \|\Pi_{\phi R} - \Pi_R\|_F, \quad d_N = \|\Pi_{\phi N} - \Pi_N\|_F,$$

and

$$e_\Sigma = \frac{\|\phi \hat{\Sigma} \phi^\top - \hat{\Sigma}\|_F}{\|\hat{\Sigma}\|_F + \varepsilon}.$$

For conformal geometry compute

$$e_{[\Sigma]} = \min_{\lambda > 0} \frac{\|\phi \hat{\Sigma} \phi^\top - \lambda \hat{\Sigma}\|_F}{\|\hat{\Sigma}\|_F + \varepsilon}.$$

Step 10. Report accept/reject outcome. A strict morphism must pass sector and covariance residual gates. A conformal morphism must pass sector gates and the conformal covariance residual gate.

Remark 5 (No CTMT estimator monopoly). The protocol deliberately does not prescribe one universal estimator for $\hat{\Sigma}$. CTMT uses the best available covariance model for the measurement protocol and then imposes resolution-relative interpretation and admissibility gates.

11 Falsification criteria

Falsifier 5 (Rank instability). If the rank of J is unstable under admissible perturbations, sampling windows, or threshold choices, then the resolved/null decomposition is not reliable for that chart.

Falsifier 6 (Null leakage). For a claimed null-supported unresolved covariance C_{null} , large persistent leakage

$$\ell = \frac{\|\Pi_R C_{\text{null}} \Pi_R\|_F + \|\Pi_R C_{\text{null}} \Pi_N\|_F}{\|C_{\text{null}}\|_F + \varepsilon}$$

falsifies the null-support hypothesis.

Falsifier 7 (Cross-block instability). If the normalized coupling

$$\kappa_{RN} = \frac{\|C_{RN}\|_F}{\|\Sigma\|_F + \varepsilon}$$

changes erratically across stabilized windows without a protocol explanation, then the covariance geometry is not stable.

Falsifier 8 (False conformal acceptance). If a transformation rescales only Σ_R or only Σ_N while leaving other covariance blocks unchanged, it is not a conformal CTMT morphism unless a separate sector-relative geometry has been explicitly declared and tested.

Falsifier 9 (Fisher-only false positive). If two charts have similar Fisher pullbacks but significantly different

$$N, \quad \Sigma_N, \quad C_{RN},$$

then Fisher similarity is a false positive for CTMT covariance-geometry similarity.

Falsifier 10 (Transport non-composability). If covariance geometry appears preserved on individual chart transitions but fails under composition around a closed loop, then the local similarities do not define admissible CTMT transport.

12 Limitations and non-claims

Non-claim 26 (No new universal covariance estimator). This paper does not claim that CTMT replaces established covariance estimation. It provides a geometric structure for interpreting and testing covariance once it is estimated or declared.

Non-claim 27 (Jacobian does not determine full covariance). The Jacobian determines the resolved sector and null support. It does not determine the complete covariance spectrum. Full covariance requires data, sensor-noise models, residual second moments, or another declared statistical source.

Non-claim 28 (No Fisher-only closure). The Fisher pullback

$$G_F = J^\top \Sigma^{-1} J$$

is a useful derived diagnostic, but it is not sufficient to determine CTMT covariance geometry.

Non-claim 29 (No automatic global theorem). Local covariance-geometry compatibility does not by itself prove global bundle consistency, nonlinear phase-Hessian lift consistency, or holonomy closure. Those are separate transport problems.

Non-claim 30 (No physical universality claim). This paper does not claim a final physical equation, earthquake prediction scheme, Navier–Stokes solution, quantum measurement derivation, or universal laboratory validation. It states a local covariance-geometry structure and its falsification criteria.

Non-claim 31 (No sector-relative shortcut). Sector-relative covariance equivalences may be useful in specialized protocols, but they are not part of the default CTMT covariance geometry. They require their own axioms, diagnostics, and falsification batteries.

13 Discussion

The resolution-geometry formulation of CTMT identifies the local morphism problem with automorphisms of

$$\mathcal{G} = (O; R, N, \Sigma).$$

This paper clarifies the internal role of the covariance component. The covariance is not merely a noise matrix and not merely a Fisher weight. Once a resolved/null decomposition is fixed, covariance becomes a structured geometric object:

$$\Sigma \longleftrightarrow (\Sigma_R, \Sigma_N, C_{RN}).$$

The null block Σ_N records covariance invisible to first-order resolved parameter motion. The cross block C_{RN} records coupling between resolved and unresolved observational directions. This coupling is not optional: it changes the stabilizer group and therefore changes the admissible morphism class. Systems with negligible C_{RN} admit a block-product automorphism group. Systems with non-negligible C_{RN} admit only the stabilizer of that coupling.

The resulting hierarchy is:

$$\begin{aligned} \text{data/protocol} &\longrightarrow \hat{\Sigma}, \\ J &\longrightarrow (R, N), \\ (R, N, \hat{\Sigma}) &\longrightarrow (\hat{\Sigma}_R, \hat{\Sigma}_N, \hat{C}_{RN}), \\ (R, N, \hat{\Sigma}_R, \hat{\Sigma}_N, \hat{C}_{RN}) &\longrightarrow \text{Aut}(\mathcal{G}). \end{aligned}$$

This hierarchy avoids two overstatements. First, it avoids saying that CTMT has a universal covariance estimator. Second, it avoids saying that the Jacobian alone determines full covariance. The conservative statement is stronger and more precise: CTMT determines how covariance must be decomposed, transported, compared, and falsified relative to resolution geometry.

14 Conclusion

The final conservative statement of CTMT covariance geometry is:

$$\text{CovGeom}(J, \Sigma) = (R, N, \Sigma_R, \Sigma_N, C_{RN}),$$

where

$$R = \text{Im } J, \quad N = \ker J^\top,$$

and

$$\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^\top & \Sigma_N \end{pmatrix}$$

relative to $O = R \oplus N$.

Strict admissible morphisms preserve this geometry exactly. Conformal admissible morphisms preserve it up to one common positive covariance scale, but only when the protocol explicitly discards absolute scale. Null-supported unresolved covariance is constrained by the canonical projector

$$\Pi_N = I - JJ^+,$$

but the full covariance spectrum must be supplied by data or a declared statistical model.

Therefore CTMT does not need to claim ownership of covariance estimation. Its contribution is the resolution-relative covariance geometry that determines similarity, admissibility, automorphism stabilizers, Fisher insufficiency, and falsification gates.

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